

WORKFLOW EXPLANATION DOCUMENT

*Automated Construction Management System
Utilizing AI Agent “Nong Nai Chang Khon Foo”*

Project Members

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1. System Overview

The “Nong Nai Chang Khon Foo” system is a fully automated, AI-powered construction management platform that eliminates manual site reporting bottlenecks by transforming the LINE messaging application into an intelligent data ingestion interface. Rather than relying on traditional textual reports or periodic manual updates, this system enables site foremen to submit their daily progress reports conversationally through a LINE Official Account chatbot. The AI backbone then performs entity extraction, schedule comparison, and strategic analysis—all in real time.

The system is architected around n8n, a low-code workflow automation platform, which acts as the central Workflow Orchestrator. n8n coordinates the seamless exchange of data between four core external services: LINE Messaging API (the user interface for site personnel), Google Gemini AI (the cognitive reasoning engine), Google Sheets (the centralized Master Plan database), and Gmail (the automated executive reporting channel).

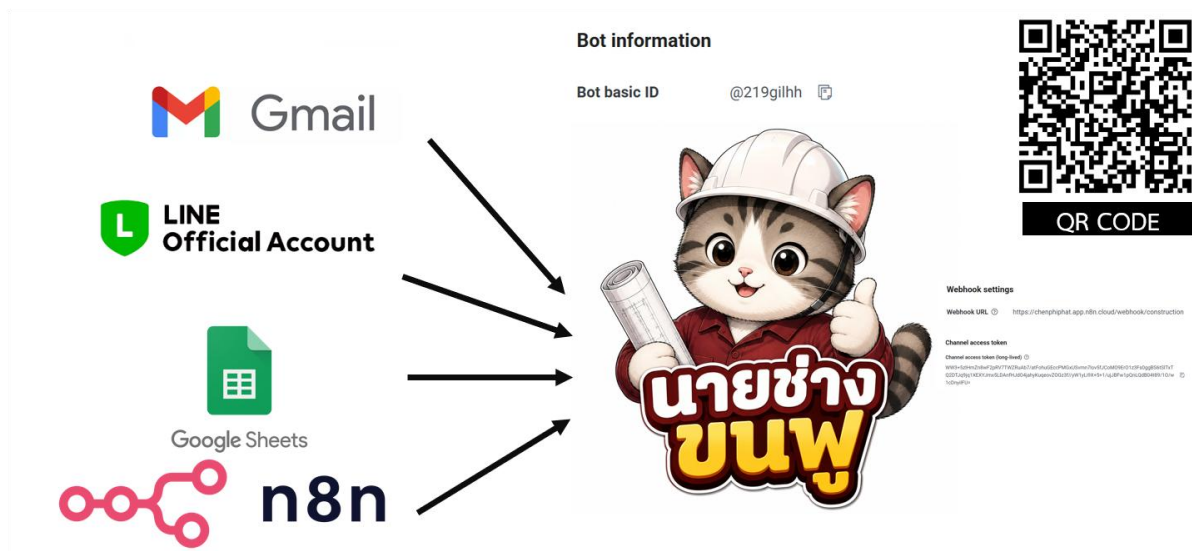


Figure 1: System overview and mascot of the Nong Nai Chang Khon Foo platform

1.1 Core Objectives

- **Eliminate “Data Lag”:** Transition site-to-office reporting from weekly batch summaries to real-time continuous updates.
- **Enforce Data Integrity:** Validate every incoming report before committing data to the Master Plan database.
- **Enable AI-Powered Analysis:** Use LLM-based reasoning to compare reported progress against the Critical Path Method (CPM) baseline and generate actionable mitigation strategies.
- **Automate Executive Notifications:** Deliver a structured HTML status report to the Project Engineer via Gmail after every valid update.

1.2 Technologies and Platforms

Technology Stack	
Orchestrator	n8n (cloud-hosted, Low-code Workflow Automation)
User Interface	LINE Official Account — Bot ID: @219gilhh
AI / LLM Engine	Google Gemini (models/gemini-2.5-flash-preview) via Google Generative AI API
Database	Google Sheets (Master Plan spreadsheet)
Notification	Gmail API (Gmail OAuth2 — automated HTML email to Project Engineer)
Entry Point	Webhook URL: https://chenphiphat.app.n8n.cloud/webhook/construction

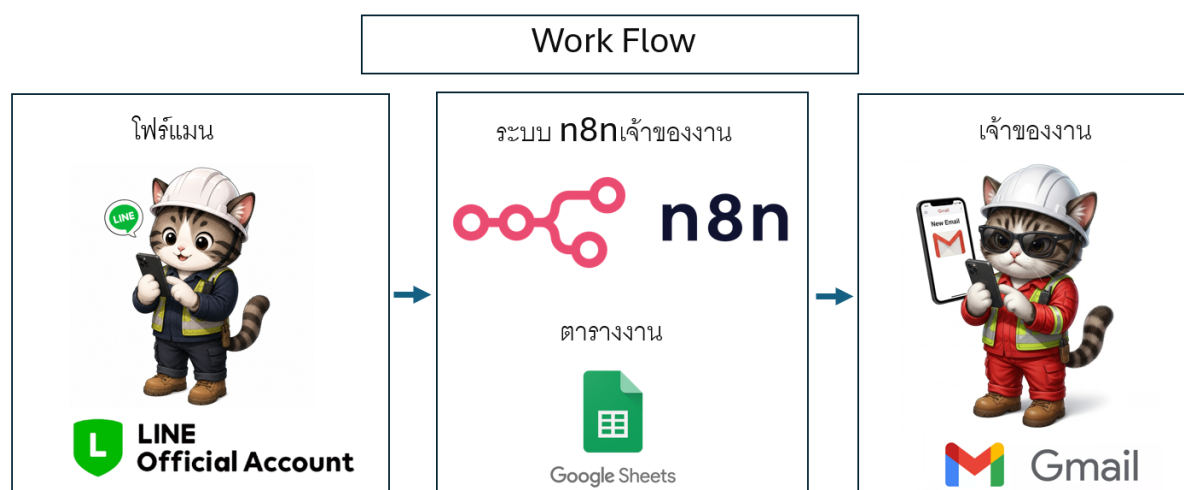


Figure 2: High-level data flow from site foreman (LINE) through n8n to Google Sheets and Gmail

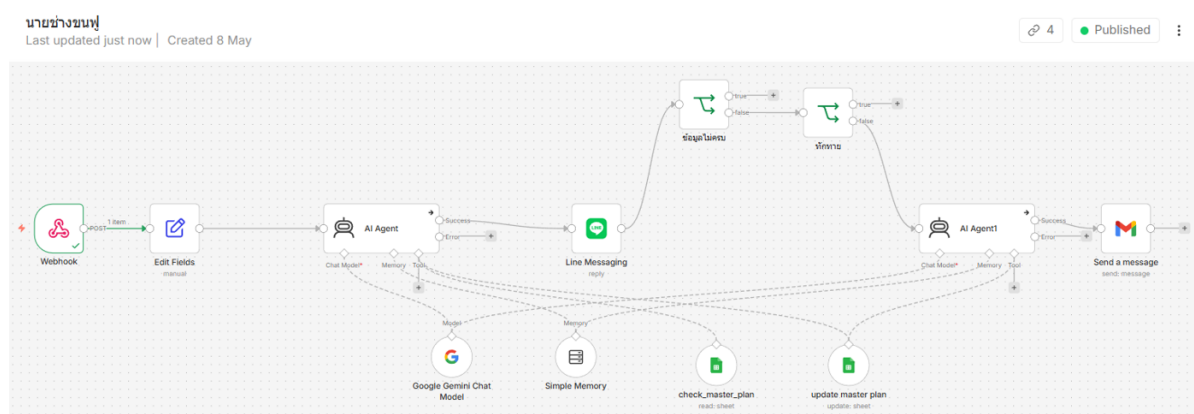


Figure 3: Complete n8n workflow canvas showing all connected nodes

2. Workflow Components Explanation

The n8n workflow is structured as a linear pipeline consisting of seven primary nodes and two conditional branches. Each node fulfills a specific role in the data lifecycle—from initial message reception to final executive notification. The following subsections document each node in precise technical detail.

2.1 Node 1: Webhook — Data Ingestion Gateway

Webhook Node — Configuration Summary	
Node Type	Webhook (HTTP Trigger)
HTTP Method	POST
Endpoint Path	construction
Full URL	https://chenphiphat.app.n8n.cloud/webhook/construction
Authentication	None (managed via LINE API signature validation)
Response Mode	Immediately (returns 200 OK upon receipt)

The Webhook node serves as the single, authenticated entry point into the entire workflow. It listens for inbound HTTP POST requests originating from the LINE Platform whenever a user sends a message to the chatbot. This trigger-based mechanism ensures that workflow execution is event-driven and on-demand, consuming no resources when idle.

Upon receiving a POST request, LINE delivers a JSON payload conforming to the LINE Messaging API specification. The top-level structure of this payload is as follows:

```
{ "destination": "...", "events": [ { "type": "message", "replyToken":  
"<token>", "message": { "type": "text", "text": "<foreman input>" } } ] }
```

The two most critical values extracted at this stage are:

- **events[0].message.text** — The natural language report submitted by the foreman (e.g., “Project: Building A, Date: 1 May 2026, Piling work 50% complete”).
- **events[0].replyToken** — A one-time token required to send a reply message back to the specific LINE conversation thread.

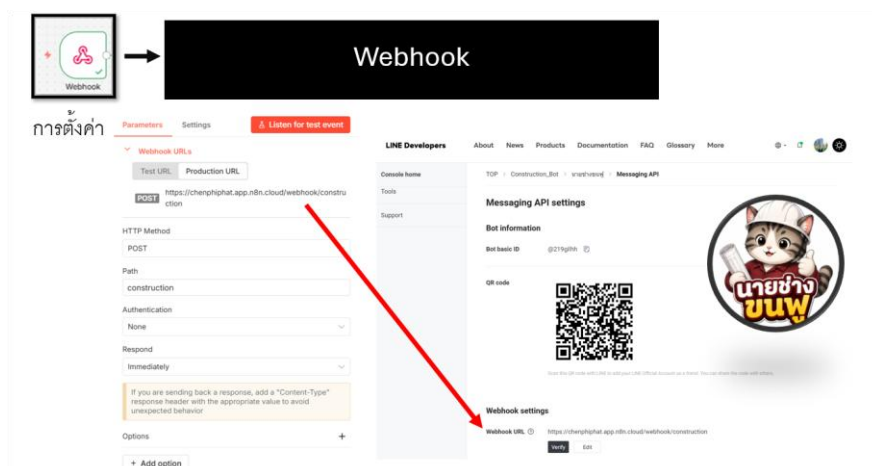


Figure 4: Webhook node configuration displaying the endpoint URL and HTTP method settings

2.2 Node 2: Edit Fields — Data Cleaning and Transformation

Edit Fields Node — Configuration Summary	
Node Type	Set / Edit Fields
Purpose	Flatten nested JSON payload from LINE into a clean, standardized key-value format
Source Field	{{ \$('Webhook').item.json.body.events[0].message.text }}
Source Token	{{ \$('Webhook').item.json.body.events[0].replyToken }}
Output Keys	message_text, reply_token

The raw LINE payload is deeply nested, with the user's message buried inside an events array at index 0. If downstream nodes attempt to access this nested path directly, parsing inconsistencies or null-reference errors can occur. The Edit Fields node resolves this by performing a Data Flattening operation: it maps the nested source paths to simple, flat variable names that all subsequent nodes can access without additional path traversal.

This node also acts as the system's first quality gate. If the payload does not contain a valid events[0] entry (for example, if LINE sends a system follow event rather than a user message), the mapping will produce empty fields, which will subsequently be caught by the validation logic in the IF node downstream.

The transformation mapping is as follows:

Output Variable Name	Source Expression (n8n Syntax)
message_text	{{ \$('Webhook').item.json.body.events[0].message.text }}
reply_token	{{ \$('Webhook').item.json.body.events[0].replyToken }}

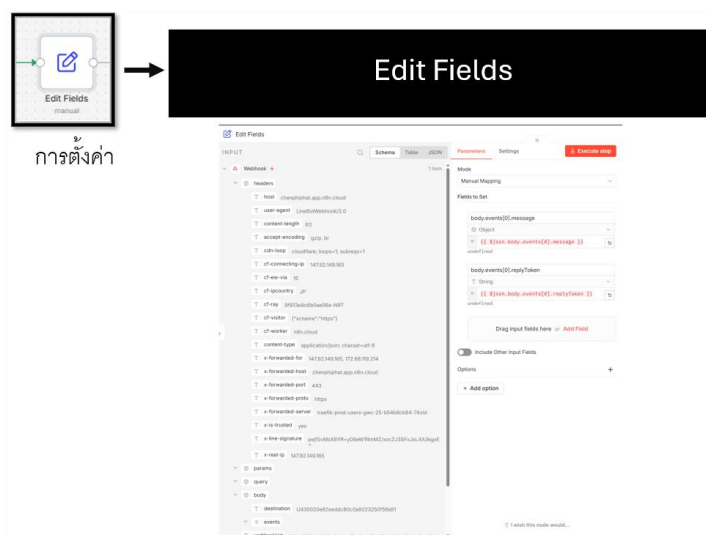


Figure 5: Edit Fields node mapping nested LINE payload fields to flat output variables

2.3 Node 3: AI Agent (Update Work) — Cognitive Core

AI Agent Node — Configuration Summary	
Node Type	AI Agent (LangChain Agent)
LLM Provider	Google Gemini via Google Generative AI API
Model	models/gemini-2.5-flash-preview
Memory	Simple Memory (in-session context retention)
Custom Tools	check_master_plan, update_master_plan
Memory Buffer	Session-scoped (clears per workflow execution)

The AI Agent node is the cognitive core of the entire system. It receives the cleaned message_text from the Edit Fields node and applies Large Language Model (LLM) reasoning to perform three critical functions sequentially: Entity Extraction, Schedule Comparison, and Strategic Report Generation.

2.3.1 Entity Extraction

Using Google Gemini, the agent parses the foreman's natural language input and extracts three mandatory data entities:

- **Project Name** — identifies which construction project is being reported on.
- **Report Date** — the date on which the work was performed (not necessarily today).
- **Completion Percentage** — the reported progress percentage for the specific task.

If any of these three entities cannot be resolved from the input, the agent does not attempt a guess. Instead, it returns a standardized error string that is subsequently routed to the incomplete data handling branch (see Section 2.4).

2.3.2 System Prompt Design

The agent operates under a carefully engineered system prompt that defines its persona, responsibilities, language behavior, and output format. The prompt instructs the agent to:

1. Act as an AI construction management assistant named “Nong Nai Chang Khon Foo.”
2. Extract project name, date, and completion percentage from user input.
3. Invoke the `check_master_plan` tool to retrieve the current CPM baseline schedule.
4. Compare reported progress against the baseline to determine schedule status (on-track, ahead, or delayed).
5. If delayed, calculate the exact delay duration and identify all dependent tasks impacted in the critical path.
6. Invoke the `update_master_plan` tool to commit the validated data to Google Sheets.
7. Generate a structured HTML executive report for the Project Engineer’s Gmail notification.
8. Reply to the foreman via LINE with a concise, friendly status acknowledgment.

2.3.3 Tool Integration

The agent is equipped with two custom tools that interface directly with the Google Sheets database:

Tool Name	Function
<code>check_master_plan</code>	Reads the current Master Plan from Google Sheets. The agent uses this data to compare the reported completion date and percentage against the CPM baseline schedule, enabling automated Gap Analysis.
<code>update_master_plan</code>	Writes the validated, extracted data (project name, date, completion %) back to the designated row in Google Sheets. This is the sole mechanism by which the database is updated, ensuring that only AI-validated data is committed.

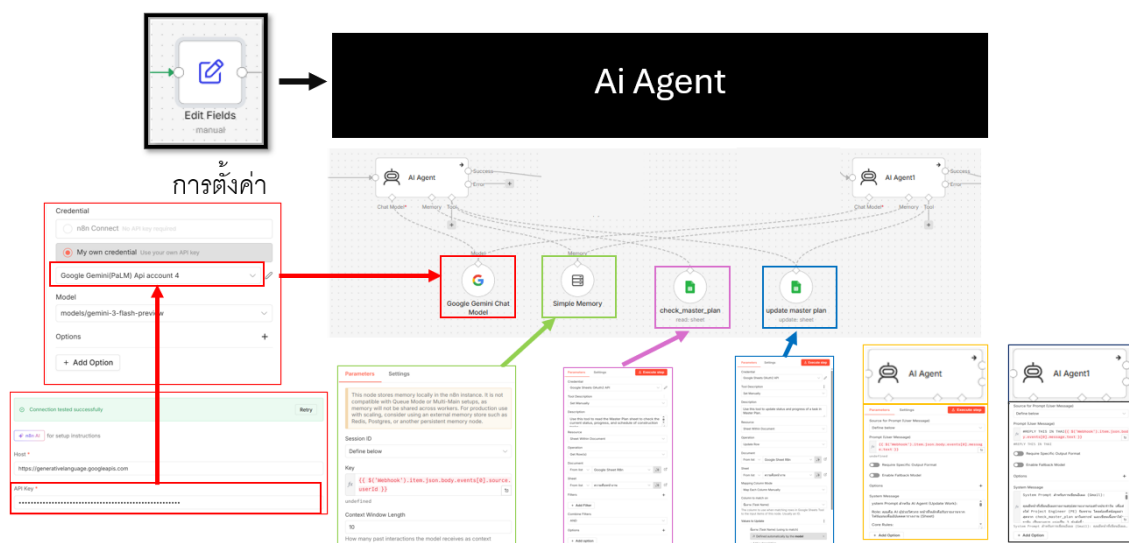


Figure 6: AI Agent node settings showing Gemini model selection, memory configuration, and connected custom tools

2.4 Node 4: IF Function — Validation Logic

IF Function Node — Configuration Summary	
Node Type	IF (Conditional Branch)
Evaluated Field	{{ \$('AI Agent').item.json.output }}
Condition A	Output equals: “ข้อมูลไม่ครบ (ไม่มี ชื่อโครงการ, โซฟ้านแมน, วันที่, และความคืบหน้าของงาน)”
Condition B	Output equals: “สำเร็จ บันทึกความคืบหน้าการทำงานเรียบร้อยแล้ว”
True Branch	Route to LINE Reply (request missing information from foreman)
False Branch	Route to Gmail (send complete report to Project Engineer)

The IF Function node implements a critical Data Hygiene Gate that intercepts the AI Agent's output and evaluates its content before allowing any data to proceed to the database or notification systems. This is essential because the AI Agent, when it cannot extract all three required entities, returns a predefined incomplete-data error string rather than a partial record.

The conditional logic operates on two possible string values returned by the AI Agent:

- **Incomplete Data String:** If the output matches the error string, the IF node routes the workflow to the LINE Reply node (Section 2.5), which sends a prompt back to the foreman requesting the missing fields (project name, date, or completion percentage).
- **Success String:** If the output indicates successful processing, the workflow routes to the Gmail node (Section 2.6) for executive notification, confirming that the database has been updated.

This two-branch architecture ensures that the Google Sheets database is never contaminated with partial or malformed records. The administrative burden on the Project Engineer is also reduced, as validation is fully automated without requiring human review of incoming reports.



Figure 7: IF node configuration showing the two conditional branches based on AI Agent output content

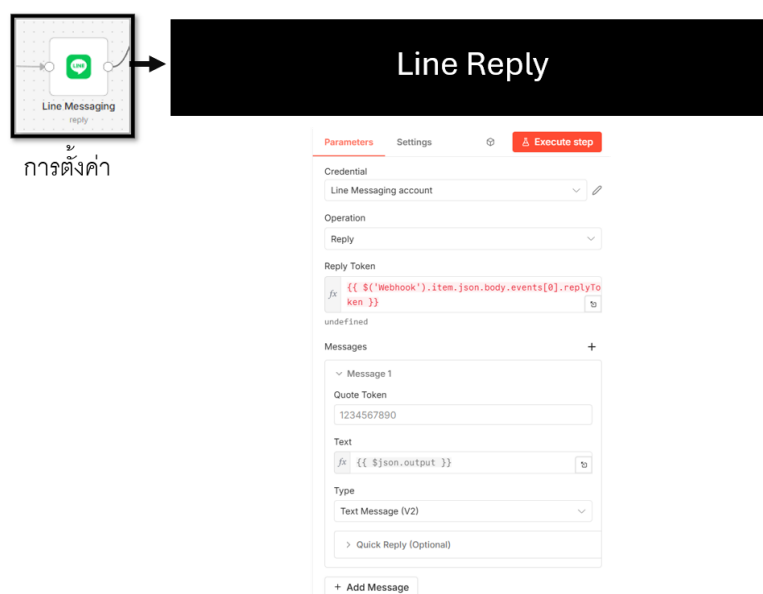


Figure 8: LINE reply node triggered when the IF validation detects incomplete foreman input

2.5 Node 5: LINE Messaging (Reply) — Incomplete Data Handler

LINE Messaging Reply Node — Configuration Summary	
Node Type	LINE Messaging (n8n integration)
Operation	Reply
Reply Token	{{ \$('Webhook').item.json.body.events[0].replyToken }}
Message Type	Text Message (V2)
Message Body	{{ \$json.output }} (AI-generated request for missing fields)

This node is activated exclusively on the “True” branch of the IF Function, meaning it fires only when the AI Agent has determined that the foreman’s input was missing one or more required entities. It uses the replyToken extracted during data ingestion to send a targeted, contextually aware reply back to the specific LINE conversation thread.

The message body is dynamically generated by the AI Agent and passed through the workflow. Rather than sending a static error message, the system provides a conversational prompt that explicitly states which fields are missing—for example: “Please provide the project name, reporting date, and work completion percentage so I can record your progress.” This UX-sensitive approach minimizes confusion and reduces the time required for the foreman to re-submit a corrected report.

Critically, this branch does NOT write any data to Google Sheets and does NOT send an email to the Project Engineer, preserving both database integrity and executive notification quality.

3. Database and Notification Integration

3.1 Google Sheets — Master Plan Database

Google Sheets Integration — Configuration Summary	
Integration Method	Native n8n Google Sheets node (OAuth2)
Operations Used	Read (check_master_plan tool), Write/Update (update_master_plan tool)
Triggered By	AI Agent custom tool calls
Data Committed	Task Name, Start Date, End Date, Status, Completion Percentage (%)

Google Sheets serves as the centralized, persistent Master Plan database for the project. It is structured as a CPM (Critical Path Method) schedule table containing one row per construction task, with columns for Task Name, Predecessor Task, Start Date, End Date, Status, and Completion Percentage.

Database operations are exclusively mediated through the AI Agent's two custom tools. The AI Agent first calls `check_master_plan` to retrieve the current baseline for comparison, then—if data validation passes—calls `update_master_plan` to commit the new completion percentage and update the task status (e.g., from "Pending" to "In Progress" or "Done"). This read-before-write pattern is essential for enabling the Gap Analysis that drives the delay detection logic.

The CPM table used during system testing contained the following baseline tasks for a residential construction project (calendar dates projected for 2026):

Task Name	Predecessor	Start	End	Status	%
Pile & Foundation	—	01/05/2026	05/05/2026	Pending	0%
Ground Beams & Columns	Pile & Foundation	06/05/2026	12/05/2026	Pending	0%
Roof Structure	Ground Beams	13/05/2026	17/05/2026	Pending	0%
Roofing	Roof Structure	18/05/2026	20/05/2026	Pending	0%
Plastering & Decoration	Roofing	21/05/2026	30/05/2026	Pending	0%
Inspection & Handover	Plastering	31/05/2026	31/05/2026	Pending	0%

ชื่องาน (Task Name)	งานที่ต้องเสร็จก่อน (Predecessor)	วันเริ่ม (Start)	วันเสร็จ (End)	สถานะ (Status)	ความคืบหน้า (%)
งานเสาเข็มและฐานราก	-	01/05/2026	05/05/2026	Pending	0%
งานคานคอดินและเสา	งานเสาเข็มและฐานราก	06/05/2026	12/05/2026	Pending	0%
งานโครงหลังคา	งานคานคอดินและเสา	13/05/2026	17/05/2026	Pending	0%
งานมุงหลังคา	งานโครงหลังคา	18/05/2026	20/05/2026	Pending	0%
งานก่ออิฐฉาบปูนและตกแต่ง	งานมุงหลังคา	21/05/2026	30/05/2026	Pending	0%
ตรวจรับและส่งมอบบ้าน	งานก่ออิฐฉาบปูนและตกแต่ง	31/05/2026	31/05/2026	Pending	0%

Figure 10: The CPM baseline table as stored in Google Sheets, used for schedule comparison during testing

3.2 Gmail API — Automated Executive Reporting

Gmail Node — Configuration Summary	
Node Type	Gmail (n8n native integration)
Authentication	Gmail OAuth2 API
Operation	Send Message
Recipient	chenphiphat.chen@gmail.com (Project Engineer)
Subject	report
Email Type	HTML
Message Body	{{ \$json.output }} (AI-generated HTML executive report)
Triggered By	False branch of IF Function (successful data submission only)

The Gmail node is the final output stage of the successful data processing branch. After the AI Agent has validated the foreman's input, extracted all required entities, compared progress against the CPM baseline, and committed the data to Google Sheets, it generates a comprehensive HTML-formatted executive report.

This report is transmitted automatically to the Project Engineer's Gmail address and contains the following structured information:

- **Summary Header:** Project name, reporting date, and foreman identifier.
- **Progress Status:** The reported task, completion percentage, and current CPM schedule position.
- **Gap Analysis Result:** Whether the project is on-track, ahead of schedule, or delayed. If delayed, the exact delay duration in calendar days is stated.
- **Critical Path Impact:** Identification of all downstream tasks affected by the delay, based on dependency mapping from the CPM table.
- **Action Plan:** Proactive, AI-generated strategic recommendations for mitigating delays—such as authorizing overtime (OT) work, reallocating labor resources, or accelerating procurement for the next task.

By delivering this analysis automatically within seconds of a foreman's report, the system shifts project management from a reactive, report-driven model to a proactive, decision-support model. The Project Engineer receives the intelligence needed to intervene before minor delays compound into critical path failures or trigger Liquidated Damages clauses.

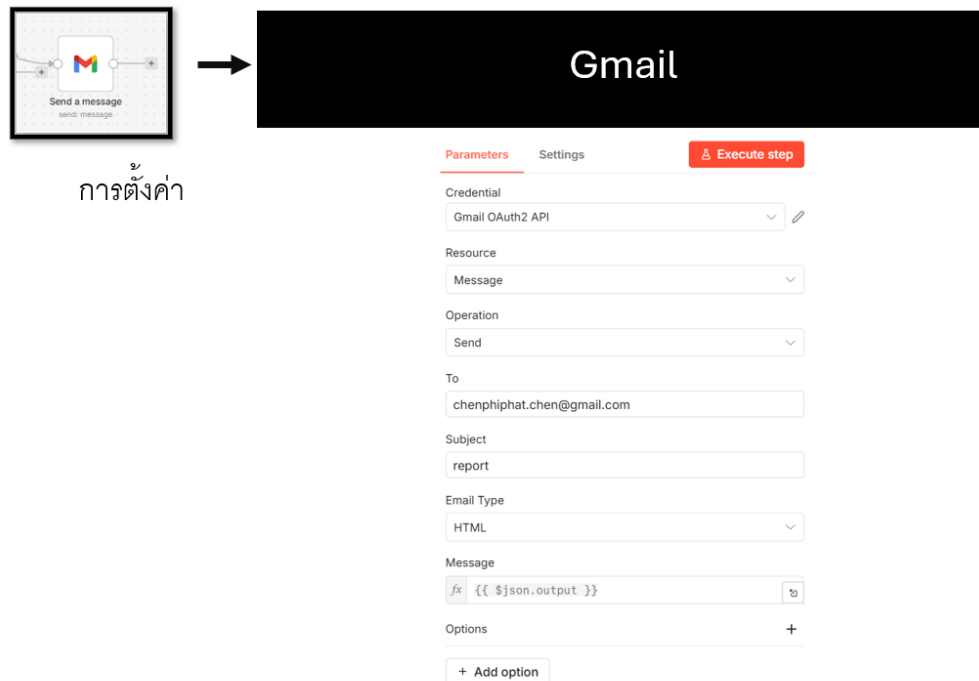


Figure 9: Gmail node configuration showing OAuth2 credentials, recipient address, and HTML message body mapping

4. Testing Results Summary

The complete system underwent structured functional testing using four distinct scenarios, each designed to validate a specific operational pathway within the workflow. All tests were conducted using simulated data projected for the 2026 construction timeline.

4.1 Scenario 1: System Initialization and Greeting

A non-reporting message ("Hi") was sent to the LINE bot to verify conversational handling. The AI Agent correctly classified the input as a non-data submission and responded with a friendly prompt guiding the user to submit a proper work report. No data was written to Google Sheets and no email was dispatched, confirming that the system correctly discriminates between conversational and reporting intents.

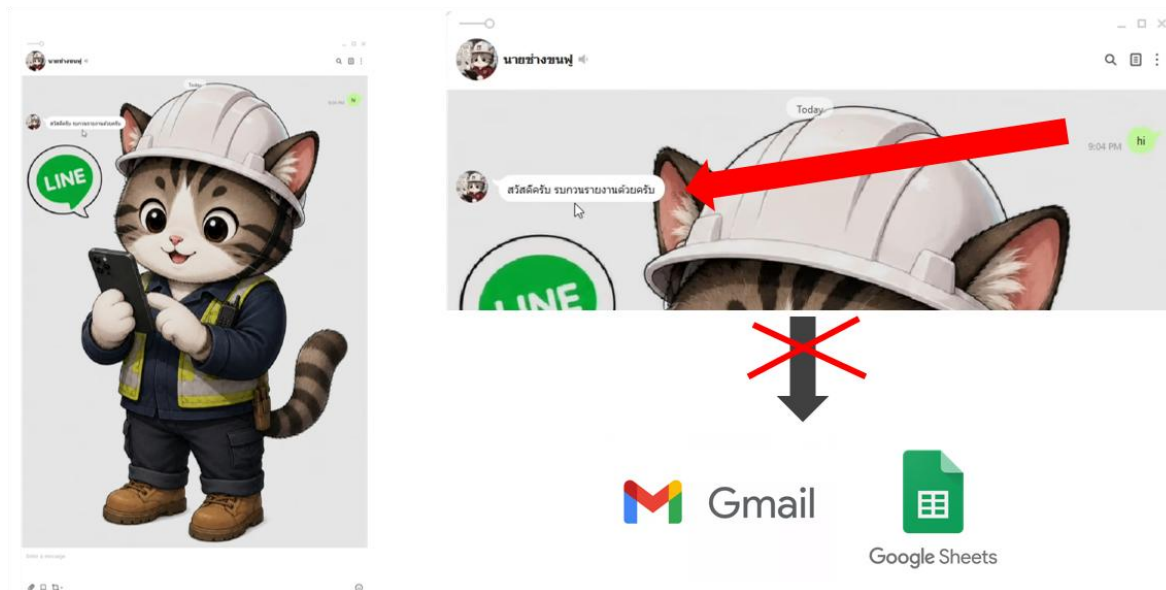


Figure 12: System response when a greeting message is detected—no database update or email is triggered

4.2 Scenario 2: Nominal Reporting (Ahead of Schedule)

The foreman submitted a report indicating 50% completion of the Pile & Foundation task on Day 1 of an allocated 5-day duration. The AI Agent invoked `check_master_plan`, determined the project was significantly ahead of the baseline schedule, and updated the Master Plan via `update_master_plan`. The LINE bot acknowledged the update with a positive status message, and the Project Engineer received an HTML email confirming the ahead-of-schedule status.

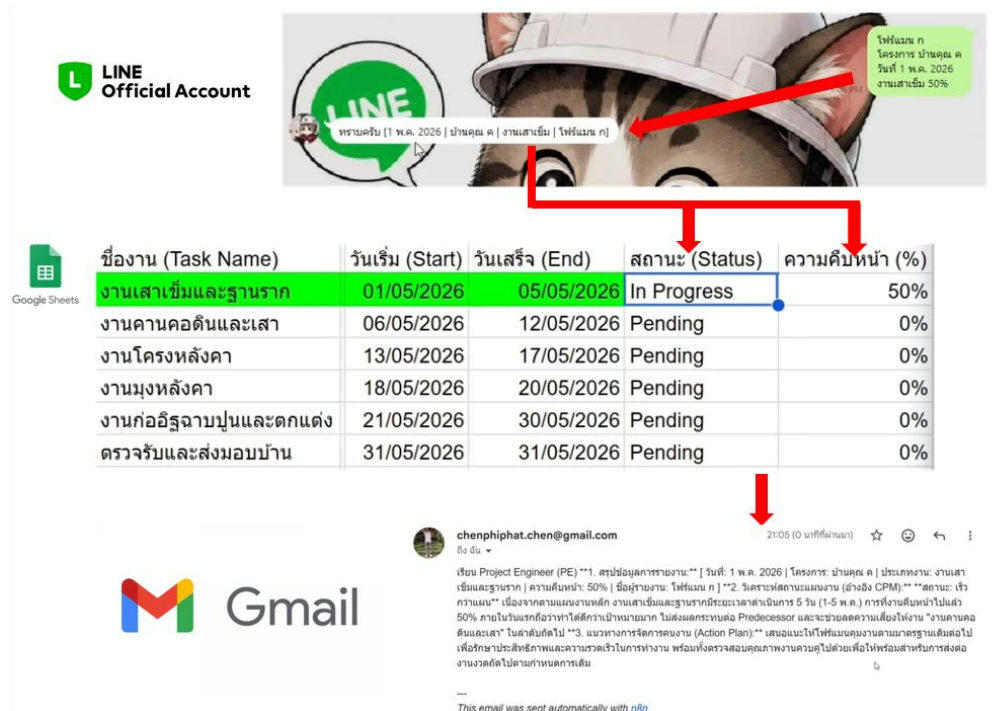


Figure 13: Successful work update showing Google Sheets database change and Gmail executive report delivery

4.3 Scenario 3: Delayed Reporting (Critical Path Impact)

The foreman reported completion of the piling work on May 7, 2026, two calendar days after the CPM mandated completion date of May 5. The AI Agent's Gap Analysis identified the 2-day delay and mapped its cascading impact on the successor task (Ground Beams & Columns), which was now unable to commence on its planned start date. The executive email report included a detailed Action Plan recommending overtime authorization and expedited procurement to recover the schedule. This scenario validated the system's core value proposition: automated delay detection with proactive mitigation strategy generation.

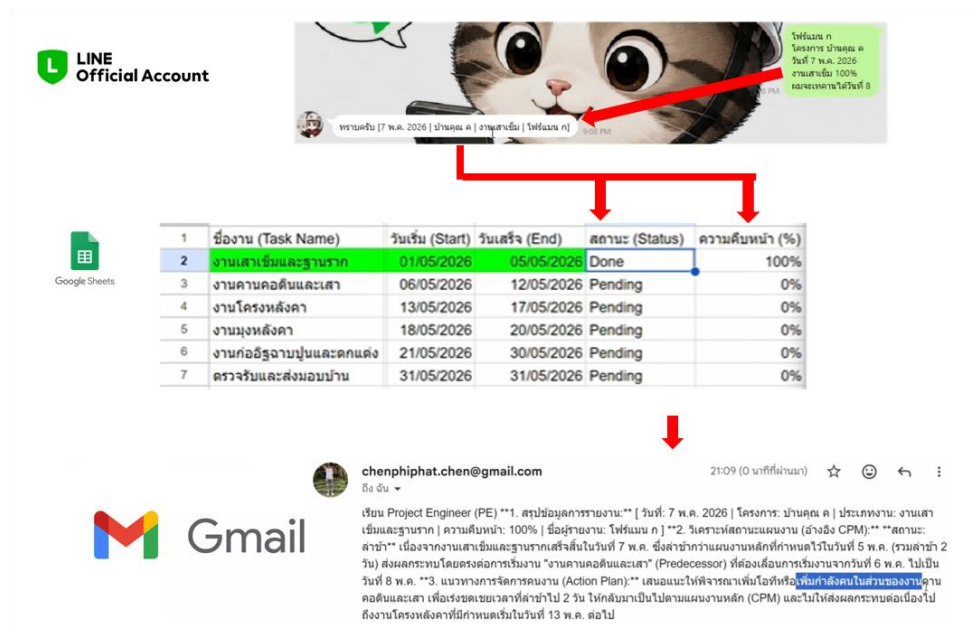


Figure 14: Delayed reporting scenario—Google Sheets updated to 100% Done, Gmail report contains delay analysis and recommended action plan

4.4 Scenario 4: Incomplete Data (Validation Logic Trigger)

A deliberately malformed report was submitted omitting the project name and specific date. The IF Function node successfully intercepted the AI Agent's incomplete-data output string and routed the workflow to the LINE Reply branch. The foreman received a contextual prompt explicitly listing the missing fields required. No write operation was performed on Google Sheets and no email was sent to the Project Engineer, demonstrating the robustness of the data hygiene mechanism.

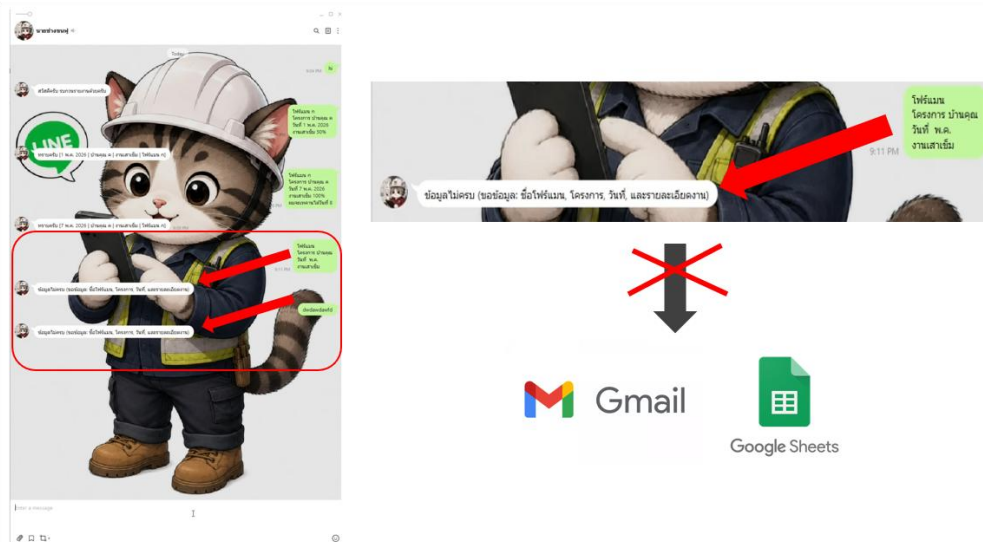


Figure 15: Incomplete data handling—the system prompts the foreman for missing fields without updating any backend systems

5. Final Summary

The “Nong Nai Chang Khon Foo” workflow represents a comprehensive implementation of AI-integrated construction management, engineered to solve a specific and measurable real-world problem: the Data Lag between site operations and project management decision-making.

The seven-node n8n workflow achieves its objectives through a clearly defined division of responsibilities. The Webhook and Edit Fields nodes form the data ingestion layer, ensuring that raw LINE payloads are received and normalized into a machine-processable format. The AI Agent constitutes the analytical layer, where Google Gemini performs entity extraction, schedule comparison, and report generation in a single, orchestrated reasoning cycle. The IF Function enforces a data hygiene layer that prevents incomplete records from corrupting the Master Plan database. Finally, Google Sheets and Gmail form the output and persistence layer, providing both a reliable data store and an automated executive reporting channel.

The four testing scenarios comprehensively validated every logical pathway within the system: nominal reporting, ahead-of-schedule detection, delay identification with cascading impact analysis, and incomplete data handling. In all cases, the system responded correctly and automatically, without requiring any manual intervention from the Project Engineer.

By transitioning construction project management from periodic, manual reporting to a real-time, AI-powered pipeline, this system delivers three categories of measurable value: Time Efficiency (elimination of Data Lag), Data Quality (automated validation and structured database updates), and Strategic Intelligence (proactive delay detection and AI-generated action plans). The architecture is also inherently scalable—additional project tasks, new analytical tools, or expanded notification channels can be integrated into the n8n canvas without restructuring the core workflow logic.